Citable by Construction

An Evidence Model for Machine-Readable Comparative Law: Three Orthogonal Axes, an Enforced Citable Subset, and Computed-versus-Authored Reconciliation

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ogy paper: it abstracts the evidence model, citability discipline, and computed-versus-authored reconciliation developed for a twelve-jurisdiction cross-border stablecoin register into a reusable method for any machine-readable comparative-law knowledge base. The worked instance is the Cross-Border Stablecoin Register (CBSR, v0.9.8); the counts and worked examples are drawn from its build at that version, but the method is domain-general. The paper is descriptive of the method, not of any jurisdiction's law; the substantive legal facts are carried in the companion working papers. **On references:** consistent with the verification discipline this paper argues for, the external entries in the References section have been verified against their primary records (publisher and standards-body pages), and each carries its DOI or standard identifier with confirmed volume, pages, and date. The paper applies its own standard to its own bibliography.

WORKING PAPER

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Abstract

Machine-readable knowledge bases of comparative regulation are increasingly consumed not by readers who can weigh a caveat but by software, retrieval pipelines, language-model agents, automated compliance tools, that will treat whatever a field asserts as fact. In that setting three distinctions that a careful human reader keeps implicit must be made explicit, structured, and enforced, or the knowledge base will present a market report as binding law, a draft provision as in force, or an unverified transcription as confirmed. This paper sets out an evidence model that makes those three distinctions first-class and orthogonal: `claim_class`, the kind of claim (a proposition of law versus a market or operational fact); `evidence_tier`, the provenance strength (confirmed against the official text, partly confirmed, or practitioner-corroborated); and `binding_status`, the binding status of the instrument the claim rests on (enacted and in force, made but not commenced, finalized policy pending legislation, a pending proposal, a prohibition, or no regime). The three axes compose without collision, and their intersection defines a lawyer-citable subset, the records a lawyer or supervisor can cite as current binding law, as the conjunction of `tier1_legal`, `in_force`, and `resolution_text`.

The paper then argues that a subset defined only by self-reported tags is not trustworthy, and sets out the build-time enforcement that makes the subset real: integrity gates that refuse a record asserting official-text confirmation without a locator, a purity gate that keeps a market fact out of the instrument a lawyer is shown, a binding-status gate that refuses to treat a located-but-not-commenced instrument as current law, a signal-provenance gate that forbids any computed result to rest on an operational fact, and a ledger cross-check that catches drift, each demonstrated to bite by a negative test that breaks it on a throwaway copy.

A second instrument, computed-versus-authored reconciliation, derives the knowledge base's downstream conclusions from a rule engine and diffs them against the hand-authored analysis, treating disagreements as findings and shipping the derived layer as a labelled preview rather than as asserted truth. A third discipline, the honest residual, publishes the backlog of not-yet-verified records as queryable data and insists that the gates are defenses, not a cure: only primary-source verification retires the liability. The method is presented as domain-general; the cross-border stablecoin register is the demonstration, and the paper closes by separating what is portable (the three axes, the citable intersection, the enforcement gates, the computed-versus-authored stance, the backlog discipline) from what is domain-specific (the dimension set and the signal table).

Keywords: legal informatics; computational law; machine-readable regulation; legal citation; comparative financial regulation; data provenance; knowledge representation; verification; stablecoins. JEL Classification: K22; K23; C81; C88; G28; O38.

§ 1. Introduction

1.1 The problem: a machine reads what a human would caveat

A comparative-regulation reference is, in the end, a set of claims of the form (jurisdiction × instrument × dimension): in this jurisdiction, under this instrument, on this dimension, the law requires, permits, or prohibits the following. A human reader of such a reference brings a great deal of unwritten judgment to each claim. She knows, without being told, that a sentence reporting that a product launched is a different kind of statement from a sentence stating what a statute requires; that a provision in a bill not yet enacted is not yet the law; and that a confident summary corroborated by a practitioner note is not the same as the official text read line by line. She reads the caveats, weighs them, and cites accordingly.

A machine brings none of that. A retrieval pipeline that surfaces a record, a language-model agent that answers a question from it, or an automated compliance tool that branches on it will treat the field it reads as the fact it needs, at whatever confidence the field' s prose happens to convey. If the reference does not record, in structure, that a given claim is a market report rather than a proposition of law, the machine cannot avoid presenting the market report as law. If it does not record that an instrument is a pending proposal, the machine cannot avoid treating the proposal as in force. If it does not record that a claim was corroborated by a summary rather than confirmed against the official text, the machine cannot avoid citing the summary as the source. The judgment the human reader supplied tacitly must, for a machine-readable reference, be supplied explicitly, encoded in fields, enforced at build time, and exposed to the consumer, or it is lost.

1.2 The contribution

This paper sets out the evidence model and the surrounding discipline that supply that judgment in structure. It has three components, each of which is the answer to one failure mode of a machine-readable legal reference.

The first component is a three-axis evidence model (§ 2). Three distinctions that a human reader keeps implicit are made first-class, required, per-record, and orthogonal: the kind of claim, its provenance strength, and the binding status of the instrument it rests on. Their intersection defines, mechanically, the subset a lawyer or supervisor can cite as current binding law (§ 3).

The second component is enforcement (§ 4). A subset defined only by self-reported tags is a promise, not a guarantee; the paper sets out the build-time gates that make the citable subset, the claim-kind separation, and the provenance discipline real rather than aspirational, and demonstrates each by a negative test that breaks the gate on a throwaway copy and confirms the build fails. The same enforcement extends to a citation firewall: no tool generates or infers a citation, and every pinpoint is verified by a human against the primary instrument, a discipline whose importance rises, not falls, when authoring is assisted by language models that will fabricate a plausible citation if permitted to.

The third component is computed-versus-authored reconciliation and the honest residual (§ § 5–6). Where a knowledge base derives downstream conclusions, those conclusions are produced by a rule engine and diffed against the hand-authored analysis; disagreements are surfaced as findings, and the derived layer ships as a labelled preview, not asserted truth. And the backlog of not-yet-verified records is published as queryable data, with the standing acknowledgment that the gates are defenses against misuse but do not themselves confirm a single claim, only primary-source verification does.

A word on what kind of contribution this is, because it is easy to mistake. The contribution is conceptual: a way of representing legal knowledge such that citability is a machine-checkable property of a record, a property the build can decide, rather than a judgment a human must remake on every read. It rests on the claim that the three distinctions above are mutually orthogonal and that their intersection is exactly "current binding law." The build-time machinery of § 4 is the carrier of that idea, not the idea itself: schema validation, the gates, and continuous integration are how the property is made to hold, but a reader who came away thinking the paper's contribution is "a continuous-integration pipeline for a legal dataset" would have the emphasis backwards. The headline is the three-axis claim typology and machine-checkable citability; the pipeline is merely its enforcement, and §§ 4 and 8 should be read in that light.

1.3 The demonstration and the scope

The method is demonstrated on a real instance: the Cross-Border Stablecoin Register, an open, versioned, schema-validated, DOI-archived, agent-queryable register of how twelve jurisdictions regulate stablecoins across fifteen dimensions, at version 0.9.8. The counts in this paper, one hundred fifty-two records, a citable subset of forty-six, an evidence-tier distribution, a declared backlog, are drawn from that build, and the worked examples (Switzerland's six citable records, Taiwan's one) are taken from its verification passes. But the paper's claims are about the method, not about stablecoin law, and § 7 separates explicitly what is portable to any comparative-regulation reference from what is specific to this domain. The paper is descriptive of a methodology; it advocates no regulatory outcome and describes no non-public arrangement.

§ 2. The three-axis evidence model

2.1 First axis: `claim_class`: the kind of claim

The first and most easily overlooked distinction is between a proposition of law and a market or operational fact. A record that states what a binding instrument requires, permits, or prohibits is a proposition of law: it can be cited to the instrument, and its truth is a matter of what the instrument says. A record that states what is actually live, which product launched, who is registered, what rails or liquidity exist, is a market or operational fact: it is true as of a date, and its truth is a matter of the state of the world, not of any instrument's text. These are different kinds of claim, and conflating them is the most consequential error a legal reference can make, because a well-sourced market fact can be mistaken for citable law precisely when it is most confidently stated.

The model makes the distinction a required, per-record field with two values. `tier1_legal` is a proposition of law; `tier2_operational` is a market or operational fact. Three properties make the field load-bearing rather than cosmetic. It is orthogonal to the other two axes, so a draft statute (a legal claim about an instrument not yet in force) and a confirmed launch (a well-sourced fact about the market) are represented without collision. It is per-record, not derived from the dimension: although in a typical reference one dimension is operational by construction and the rest state propositions of law, storing the field on each record is what allows an operational claim that happens to sit inside a nominally-legal dimension to be tagged for what it is, rather than silently inheriting "legal" from its dimension. And it is audited, not assumed: every record that a market-or-media-source heuristic flags is read by hand to confirm its kind, so that a record which merely cites a market announcement as corroboration for a proposition of law stays `tier1_legal`, while a record whose load-bearing fact is the market datum is tagged `tier2_operational`.

2.2 Second axis: `evidence_tier`: provenance strength

The second distinction is between knowing a claim well and knowing it less well. A proposition of law confirmed against the official text is more strongly sourced than the same proposition corroborated only by a practitioner summary, and a reference that cannot represent that difference cannot tell a consumer which of its claims rest on the source itself and which rest on a secondary account of it.

The model makes provenance strength a required field with an ordered set of values. `resolution_text` denotes a claim confirmed against the official text; `mixed` denotes a claim whose core point is confirmed against the official text but some operational detail is pending; `firm_summary` denotes a claim corroborated by law-firm or practitioner analysis but not yet checked against the official text; and a residual `unset` value marks records that predate the field and have not yet been assigned a tier. This axis is provenance, and only provenance: it says how well the claim is sourced, not what kind of claim it is or whether the instrument is in force. A confirmed product launch is well-sourced (`resolution_text`) and yet is not law (`tier2_operational`); a draft statutory provision is a legal claim (`tier1_legal`) and yet not binding (its status is `proposed`). The independence of the axes is the point.

2.3 Third axis: `binding_status`: the binding status of the instrument

The third distinction is the one that an external primary-source verification pass surfaced as a necessary correction to the second, and it is the subtlest. Finding the official text is not enough to make a claim citable as current law. An instrument may be located, read, and confirmed, and still not be in force, because it is made but not yet commenced, or finalized as policy but not yet legislated, or a bill not yet enacted. A reference that treated "the official text was located and confirmed" as equivalent to "this is current binding law" would promote a regime that does not yet bind to the citable tier on the strength of having read a statute that has not commenced.

The model therefore adds a third, orthogonal field recording the binding status of the instrument the claim rests on, distinct from how well the claim is sourced and from what kind of claim it is. Its values are `in_force_enacted` (an enacted instrument in force now), `made_not_commenced` (made or passed but not yet commenced), `finalized_policy_pending` (finalized policy not yet legislated), `pending_proposal` (a bill, a notice of proposed rulemaking, or a consultation), `prohibition` and `no_regime` (no positive permitted-activity rule to cite). The axis is what keeps the citable subset honest even after the official text is found: a regime whose text has been read but which commences in the future is recorded with its binding status, and the enforcement of § 4 refuses to treat it as current law until it is in force.

2.4 Orthogonality, and the combination space

The three axes are independent: each answers a different question, and any combination of their values is meaningful. The kind of claim (legal or operational) is independent of how well it is sourced (confirmed, mixed, or corroborated), which is independent of whether the instrument it rests on is in force (enacted, made-not-commenced, finalized-policy, proposal, or no-regime). The independence is what lets the model represent, without ambiguity, the cases that a single axis would conflate: a draft statute is `tier1_legal` (a legal claim) at `firm_summary` or `mixed` (sourced to a summary or partly to the text) with binding status `pending_proposal` (not enacted); a confirmed launch is `tier2_operational` (a market fact) at `resolution_text` (well-sourced) with no legal binding status at all; an in-force statute confirmed against its text is `tier1_legal` + `resolution_text` + `in_force_enacted`, which is the only combination that is citable as current binding law.

The value strings of the three axes are deliberately disjoint, so no filter can confuse one axis for another. In the demonstration instance the combination space is populated as follows: of one hundred fifty-two records, one hundred forty-three are `tier1_legal` and nine are `tier2_operational`; the provenance distribution is forty-eight `resolution_text`, fourteen `mixed`, sixty-three `firm_summary`, and twenty-seven `unset`; and one hundred of the one hundred fifty-two carry a populated official source locator. The two-axis count matrix of claim-kind against provenance, forty-six `tier1_legal` records at `resolution_text`, two `tier2_operational` records at `resolution_text`, and so on, is published with the build, so a consumer can see the population of every cell of the combination space instead of trusting a single headline number.

§ 3. The lawyer-citable subset

3.1 The intersection

The single most valuable question a lawyer or supervisor asks of a legal reference is: show me only what I could cite, today, as current binding law. The three-axis model answers it mechanically, as the intersection of one value on each axis:

```
claim_class == tier1_legal   AND   status == in_force   AND   evidence_tier == resolution_text
```

A record in this intersection is a proposition of law (so it is the kind of thing one cites to an instrument), about an instrument in force (so it binds now), confirmed against the official text (so the citation rests on the source itself). In the demonstration instance the intersection is forty-six records. Each is projected to exactly what a citation needs, the instrument, the pinpoint, the official source locator, and the date last reviewed, and published three ways: precomputed in the compiled dataset, so consumers never re-derive it; exposed by a dedicated query tool that returns the subset with optional jurisdiction and dimension filters; and surfaced by a "citable law only" toggle in the reference's interactive view.

A precision is owed here, because the formula names `status` as its third explicit condition while § 2 introduced `binding_status` as the third axis, and the two are distinct fields whose relationship is the point. `status` is the four-value field recording whether the claim's regime is in force (in force, transitional, proposed, or consultation); `binding_status` is the field recording the binding state of the instrument the claim rests on (enacted-and-in-force, made-not-commenced, finalized-policy-pending, a pending proposal, a prohibition, or no regime). The third axis does not appear in the citable formula as a fourth literal conjunct; it enters the subset through the provenance axis, by an enforced cap rather than a second condition. The build (§ 4.2) requires that any record at `evidence_tier == resolution_text` also carry `binding_status == in_force_enacted`, so `binding_status` governs which records may reach *`resolution_text`* at all. A located-but-not-commenced instrument is therefore excluded from the citable subset not by the `status` test acting alone but by being barred from the top provenance tier by the binding-status gate, which is precisely why the subset stays honest even after an instrument's official text has been read and confirmed. The three fields that appear explicitly in the citable formula are thus `claim_class`, `status`, and `evidence_tier`; the third axis, `binding_status`, is enforced into the subset by capping the provenance axis, and the architecture is three orthogonal axes of which two are tested directly in the filter and one is tested by the gate that caps the filter's provenance term.

3.2 Why each axis excludes something

The intersection is valuable precisely because each of its three conditions excludes a different category of true-but-not-citable record. The claim-kind condition excludes operational facts even when they are well-sourced: a confirmed product launch is a true, `resolution_text`-grade fact about the market, and it is deliberately excluded from citable law, because it is not a proposition of law. The status condition excludes draft provisions even when their text has been read: a bill's provisions are legal claims, but they do not bind, and a lawyer asking for current binding law does not want them. The provenance condition excludes unverified legal points even when the instrument is in force: a proposition of law about an in-force statute that has been corroborated only by a summary is excluded until it is confirmed against the official text. Three conditions, three categories of exclusion, each corresponding to one of the three ways a record can be true without being citable as current binding law.

The reconciliation in the demonstration instance shows the axes doing this work. Of the forty-eight records at `resolution_text`, forty-six are `tier1_legal` and citable, and two are `tier2_operational`, well-sourced market facts deliberately excluded from citable law despite their strong provenance. That exclusion is the claim-kind axis doing its job: a confirmed launch is not law, however well it is known. The worked jurisdiction examples make the same point at the status axis: a jurisdiction with a live, well-verified regime returns several citable records, while a pre-regime jurisdiction returns only its in-force layer, its draft issuance statute is excluded by status, exactly as a lawyer would want, even though the draft's text has been read.

3.3 The subset is strict by design

The citable filter is deliberately strict on the provenance axis: it admits only `resolution_text`, not `mixed`. A looser "citable with a caveat" view that included partly-confirmed records is intentionally not shipped, because a claim a lawyer is invited to cite as current binding law should rest on the official text, full stop. The strictness is a feature: a smaller subset that is genuinely citable is more useful than a larger one that carries silent caveats, because the consumer of the subset, increasingly a machine, does not read caveats.

§ 4. Enforcement: making the subset real

The gates described below are the carrier of the conceptual claim of § § 2–3, the mechanism that makes machine-checkable citability actually hold at build time, and not a second contribution in their own right. They are set out in some detail because a discipline that is only self-reported is not a discipline; but the reader should keep the concept (citability as a decidable property) in the foreground and read the plumbing as subordinate to it.

4.1 Why a self-reported subset is not enough

Everything in § § 2–3 is, so far, a set of fields a record's author fills in. If those fields are only self-reported, the citable subset is a promise: a record could claim `resolution_text` without a locator to the official text, or a citable record's "instrument" field could bundle a market fact alongside the statute, and nothing would catch it. A subset that can silently drift away from "confirmed against the source" is not a subset a lawyer can rely on. The model therefore makes the citability discipline real with build-time gates that refuse to compile a knowledge base in which the discipline is violated, and it demonstrates each gate by a negative test, a deliberate break on a throwaway copy that must make the build fail, after which the break is reverted.

4.2 The gates

Five gates enforce the discipline; all run in continuous integration on every change, so a violation cannot be merged.

Citable integrity. A `tier1_legal` record at `resolution_text` or `mixed` asserts confirmation against the official text, so it must carry an official source locator; at `resolution_text` it must additionally carry a pinpoint. A record cannot claim to have been confirmed against the source while withholding the locator to that source. The negative test strips the locator from a citable record and confirms the build fails with a precise message; the locator is then restored.

Citable purity. A citable record's instrument and pinpoint fields may carry only the legal instrument, no product launch, no admission, no named commercial counterparty. The market illustration that contextualizes a legal claim lives in a separate, optional operational-notes field that the citable projection omits. This gate exists because a real instance found exactly the failure it prevents: a record that had been projecting a market admission as part of its citable instrument, which the gate now forbids and which was moved to the operational-notes field. The instrument a lawyer is shown never bundles a market fact.

Binding-status. A `resolution_text` record must be `in_force_enacted`. This is the gate that keeps the subset honest after the official text is found: a made-but-not-commenced regime whose statute has been read is held below `resolution_text` until it commences, and the negative test confirms that promoting a pending-proposal record to `resolution_text` fails the build.

Signal-provenance. Every per-jurisdiction signal that the knowledge base's downstream rule engine reads must rest on a `tier1_legal` record; a signal resting on a `tier2_operational` fact is a build-failing violation. This gate makes the claim-kind distinction load-bearing for the computed layer (§ 5): a derived conclusion may rest only on propositions of law, never on a market fact. The negative test flips a signal-driving record to `tier2_operational` and confirms the engine reports the violation and the build fails.

Ledger cross-check. The external verification pass is recorded in an audit ledger that captures, for each verified cell, the instrument, the binding status, the official locator, the pinpoint, the verdict, and the tier applied; the build then cross-checks every record against the ledger to catch drift. A record whose tier or locator has drifted from what the ledger records fails the build.

In the demonstration instance the full suite of structural assertions and the negative-test suite both pass, the read-only invariants hold across the built register, and every validation gate is shown to bite when deliberately broken, so the citable subset, the claim-kind separation, and the provenance discipline are guaranteed by the build rather than asserted by the author.

4.3 The citation firewall

A discipline that surrounds all of the above, and that grows more important as authoring is assisted by language models, is the citation firewall: no tool generates or infers a citation, and every pinpoint is verified by a human against the primary instrument. The evidence model records what kind of thing a source establishes and how well it is known; it never manufactures the source. This matters because a language model asked for a citation will, if permitted, produce a fluent and plausible one that does not exist, and a legal reference that allowed model-generated citations would accumulate exactly the confident fabrications it most needs to exclude. The rule is stated at the point of authoring, copy the template, cite a primary source with a pinpoint, verify it yourself, and the citable-integrity and purity gates enforce its consequences: a record cannot reach the citable subset without a human-verified locator to the official instrument, and that locator cannot be diluted by a market fact. The firewall is the reason the three-axis model can be trusted in a workflow that uses machine assistance for drafting: the structure records provenance and kind, and a human supplies the citation. Stated as a transferable rule, this is a small but, the author believes, underused contribution in its own right, call it citation purity: in a machine-readable legal reference, the locator a consumer is offered as citable law must point to the legal instrument and to nothing else, with every operational illustration quarantined in a separate, non-citable field. Citation purity is what lets such a reference be authored with machine assistance at all without the machine's fluent fabrications, or its tendency to bundle a vivid market fact into a citation, reaching the citable layer; it is the firewall's enforceable consequence, and it generalizes to any domain in which a citation must be offered as authority.

§ 5. Computed-versus-authored reconciliation

This section names what the author takes to be the method's most original element and states it in domain-general form so that it can travel beyond the demonstration instance. Call it dual-path derivation with disagreement-as-finding: any knowledge base that derives a downstream legal conclusion from primitive facts can derive that conclusion twice, once by hand, once by a small auditable rule engine running over a provenance-annotated signal table, and then treat every divergence between the two derivations as a finding to be surfaced rather than an error to be silently reconciled in favour of either path. The paradigm is transferable to any reference that composes conclusions from facts: a corridor's cross-border feasibility here; a transaction's regulatory treatment, a filing's sufficiency, or a permit's availability elsewhere. The remainder of this section develops it on the demonstration instance, which is one realization of the paradigm rather than the paradigm itself.

5.1 The second instrument

A comparative-regulation reference often carries not only node-level facts but downstream conclusions composed from them: in the demonstration instance, the cross-border feasibility of a jurisdiction pair, composed from the two jurisdictions' positions. The naive way to carry such a conclusion is to author it by hand and store it; the risk is that the stored conclusion silently drifts from the facts it should follow from, and that an error in the composition is indistinguishable, to a consumer, from a correct result. The method's second instrument addresses this: the downstream conclusions are also derived by a small, auditable rule engine from a per-jurisdiction signal table, each signal annotated with the record that justifies it, and the derived conclusions are diffed against the hand-authored ones.

5.2 Disagreement as a finding

The reconciliation's stance is that a disagreement between the computed and the authored layer is a finding, not a bug to be silently fixed in favour of either side. In the demonstration instance the engine reproduces the great majority of the authored conclusions from first principles, nine of nine directed corridors and sixty-four of sixty-six pairwise categories, and the two disagreements are surfaced, not suppressed. Both turn out to share a single cause: one layer treats a jurisdiction whose regime is adopted but not yet operative as not-yet-bridgeable, while the other treats the same pair as cleanly bridgeable; the engine's disagreement is exactly the research-valuable signal that the two layers encode a regime-in-transition differently. A reconciliation that resolved such a disagreement by overwriting one layer with the other would destroy the signal; the method preserves it as a declared finding, and the time dimension then distinguishes the part of the finding that resolves by a commencement date from the part that is a genuine, durable modelling difference.

5.3 The derived layer is a labelled preview

The derived layer ships as a labelled preview, not as asserted authoritative. This is a deliberate epistemic posture: the rule engine is a check on the authored analysis and a way to surface drift, not a replacement oracle that decides the truth. The preview is published alongside the authored facts and exposed by query tools, so a consumer can see both the authored conclusion and the computed one and can inspect the derivation; but the reference does not claim that the computed conclusion is the right one where the two differ. The signal-provenance gate of § 4 ties this instrument back to the evidence model: because every signal the engine reads must rest on a `tier1_legal` record, the computed layer is derived only from propositions of law, which is precisely the invariant a date-aware or constraint-level derivation needs in order not to produce a dangerous answer from a market fact.

§ 6. The honest residual

6.1 The gates are defenses, not a cure

The most important thing the method says about itself is that its enforcement is a set of defenses against misuse, not a cure for ignorance. The gates guarantee that a record cannot claim a tier it lacks the provenance for, that a market fact cannot masquerade as citable law, that a not-yet-commenced instrument cannot be cited as current law, and that a computed result cannot rest on an operational fact. They do not, and cannot, confirm a single legal claim against its source.

Only a primary-source verification pass does that, and such a pass requires reading the official instruments and is deliberately never fabricated.

6.2 The backlog is published, not buried

The method therefore insists that the residual, the records not yet confirmed against the official text, be published as first-class, queryable data, not concealed behind the headline counts. In the demonstration instance the residual is stated plainly: a set of records confirmed at regime-status level or transcribed but not yet read against the official text remain below the citable tier, and every computed result rests on the cells beneath it, so the honest version of any "nine of nine corridors reproduced" claim is that it is only as sound as the hand-curated signal table and the cells under it. A per-cell worklist records, for each unverified record, the instrument and pinpoint and exactly what is missing; a verification queue tracks the backlog; and the standing acknowledgment is repeated wherever the counts appear: the claim-kind axis and the citable gates are the right defenses, but they do not retire the liability, only the verification pass does.

6.3 Earned tiers

A corollary discipline is that an evidence tier is earned, not asserted: the build enforces the necessary conditions for each tier (a `resolution_text` record must carry a locator, a pinpoint, and a review date; a `mixed` record a locator and a pinpoint; a `firm_summary` record a pinpoint), so a record cannot claim a provenance strength it lacks the provenance for. And an optional verification block records how a cell was confirmed, the method, what it was checked against, and the date, so that a promotion from one tier to a higher one is traceable rather than a bare change of a field. The residual is thus not only declared but governed: a record moves up a tier only by acquiring the provenance the build requires for that tier, and the move is recorded in the audit trail.

§ 7. Generalization

7.1 What is portable

The method is presented as domain-general, and the separation of the portable from the domain-specific is explicit. Portable to any comparative-regulation knowledge base are: the three orthogonal axes (the kind of claim, its provenance strength, and the binding status of the instrument); the citable intersection (the conjunction of legal-kind, in-force-status, and official-text-provenance that defines what may be cited as current binding law); the enforcement gates (citable integrity, citable purity, binding-status, signal-provenance, and ledger cross-check), each demonstrated by a negative test; the citation firewall (no machine-generated citations; human verification against the primary instrument); the computed-versus-authored reconciliation with its disagreement-as-finding stance and its labelled-preview posture; and the honest-residual discipline (the published backlog, the earned tiers, the standing acknowledgment that the gates are defenses, not a cure). None of these depends on the subject matter; each is a response to a failure mode that any machine-readable legal reference shares.

7.2 What is domain-specific

Domain-specific are: the dimension set (the columns along which the regulation is compared, here, fifteen dimensions aligned to an eight-constraint framework, but in another domain a different decomposition); the signal table (the per-jurisdiction values the downstream rule engine reads, and the composition rules that turn them into derived conclusions); and the substantive legal content of every record. These are the parts an analyst of the domain supplies; the method is the scaffolding around them. A practitioner building a comparative reference in another regulated field, data protection, securities, environmental permitting, would author a different dimension set and a different signal table and would fill the records with that field's law, and would inherit the three axes, the citable intersection, the gates, the firewall, the reconciliation, and the residual discipline unchanged.

7.3 Packaging

A final portable element is the packaging that makes a machine-readable legal reference usable and trustworthy as software: schema validation of every record against a published schema, so that an invalid record cannot be merged; versioning with a stable citable identifier, so that a citation to the reference resolves to a specific state; archival for a persistent identifier per release, so that the version cited does not move under the reader; continuous integration that runs the validation, the invariants, and the negative tests on every change; and an agent-queryable interface, so that the reference can be consumed by the software that is increasingly its primary reader, including a dedicated tool that returns only the citable subset, so that an automated consumer asking for current binding law receives exactly that and not the well-sourced market facts and draft provisions that surround it. The packaging is what turns the evidence model from a set of fields into a reference whose discipline a downstream system can rely on without re-deriving it.

7.4 A limitation of the demonstration

The method is demonstrated on a single instance, the author's own register (n = 1), and its portability is therefore argued (in § §7.1–7.3), not yet shown across domains. The three axes, the citable intersection, the gates, and the dual-path reconciliation are presented as domain-general on structural grounds, but the paper does not exhibit a second, independent realization in another regulated field, and a reader is entitled to treat cross-domain portability as a claim awaiting demonstration rather than an established result. Building such a second instance, in data protection, securities disclosure, or environmental permitting, each of which has the same legal-versus-operational, draft-versus-in-force, and confirmed-versus-corroborated distinctions, is the natural next step and the cleanest test of the generalization §7 asserts. The author flags this instead of leaving it for a reviewer to find, in keeping with the honest-residual discipline the method itself prescribes.

§8. Positioning: relation to adjacent work

This method sits at the intersection of two readerships, and it is worth positioning it against both, because it makes a different kind of claim to each. To a legal-informatics and computational-law readership, the contribution is a representation-and-enforcement claim: a way of recording, in machine-readable structure, not only what the law is but how well and how bindingly it is known, with citability made a decidable property. To a law-and-technology and financial-regulation readership, the contribution is an instrument claim: a depth-first, primary-source, citable reference for a fast-moving regulated domain, whose discipline a practitioner or supervisor can rely on without re-deriving it. The positioning below addresses the first readership's literature directly, because it is the one this paper most often fails to engage and the one whose standards the method must meet to be taken as a contribution rather than an artifact.

Machine-readable legal sources and rule representation. There is an established body of work on representing legal materials in machine-readable form, most prominently the Akoma Ntoso XML vocabulary for legislative and judicial documents (Palmirani & Vitali 2011; OASIS LegalDocML) and, for the representation of legal rules and norms, LegalRuleML (Athan et al. 2013). This method does not compete with those representations and is in a precise sense orthogonal to them: Akoma Ntoso models a document's structure and LegalRuleML models a norm's logical content, whereas the three axes here model the epistemic status of a recorded proposition, what kind of claim it is, how well it is sourced, and whether the instrument it rests on is in force. The natural relationship is complementary: the evidence axes could in principle annotate an Akoma Ntoso fragment or a LegalRuleML rule, attaching to it the provenance and binding-status metadata this paper argues a machine consumer requires. What the method adds that those standards do not themselves supply is the enforced citable intersection, the build-time guarantee that a proposition offered as current binding law is legal-in-kind, in-force, and confirmed against the official text, and the discipline of treating that guarantee as a property the build decides rather than a convention an author observes.

Provenance models. The evidence axes are, in part, a provenance model, and there is a general standard for provenance on the web, the W3C PROV data model and ontology (Moreau & Missier 2013; Lebo, Sahoo & McGuinness 2013). The honest positioning is that this method does not introduce a new general provenance vocabulary and could, in principle, express much of its `evidence_tier` and source-locator structure in PROV terms (a record as an entity, a verification pass as an activity, a maintainer as an agent). What it contributes beyond a general provenance ontology is twofold and domain-specific: first, the legal axes that PROV does not have a reason to carry, `claim_class` (proposition-of-law versus market fact) and `binding_status` (the in-force state of the instrument), which are exactly the distinctions a legal consumer needs and a general model leaves to the application; and second, enforcement, the gates that make the provenance claims non-self-reported, which a vocabulary by itself does not provide. PROV says how to describe provenance; this method says how to make a particular, legally-load-bearing slice of it bind.

Temporal norms and the norm lifecycle. The most important engagement, because it both validates and sharpens the model, is with the AI-and-law literature on temporal normative reasoning and the lifecycle of norms (Governatori & Rotolo 2010; Palmirani 2011; Sartor 2005, among others). That literature formalizes a distinction this paper's `binding_status` axis independently re-discovered: the difference between the time a norm is enacted, the time it acquires force or efficacy, and the time it becomes applicable. The register's `made_not_commenced` value, an instrument passed and published but not yet commenced, whose text may be read but which may not be cited as current binding law, is precisely the enactment-versus-efficacy distinction that this literature has formalized for two decades. Naming the connection does two things. It lends the model domain validity: the binding-status axis is not an ad hoc field invented for one register but an operationalization of a distinction the field already regards as fundamental, which is reassurance that the axis is carved at a real joint. And it locates the novelty precisely: where the temporal-norm literature formalizes the lifecycle to support automated normative reasoning (what follows from a norm at a given time), this method operationalizes the same lifecycle as an enforced citability cap in a machine-readable reference (whether a proposition may be offered as current binding law), enforced at build time by the rule that a `resolution_text` record must be `in_force_enacted`. The contribution is not the lifecycle distinction, that is the field's, but its use as a build-decidable gate on citation.

Practical alternatives. Set against the things a practitioner might use instead of this method, three contrasts remain. The breadth-first proprietary tracker, the "every-country" survey whose value proposition is coverage and whose sourcing is opaque, is the opposite trade: this method is depth-first and makes its judgment transparent, on the view that for a reference consumed by software an unverifiable claim about a hundredth jurisdiction is worth less than a verified, citable claim about a covered one, and that opacity is precisely what a machine cannot compensate for. The scrape, assembled by automated extraction from public pages, cannot reliably infer the very distinctions the method most needs (kind of claim, binding status, official-text confirmation), which is why the records here are hand-authored and human-verified at the point of citation. And the bare legal-citation standard expresses a citation without expressing how well the cited proposition is known; this method layers provenance and binding-status onto the citation, so the consumer learns not only what is cited but whether it may be cited as current binding law.

The differentiator across all of these, the standards, the provenance model, the temporal-norm logics, and the practical tools alike, is the same and can be stated in one line: this method takes the judgment a careful human reader supplies tacitly, renders it as structure a machine can read, enforces that structure at build time, and is honest, in published data, about the boundary of what it has verified.

§ 9. Conclusion

A machine-readable reference of comparative regulation will be read by software that treats whatever a field asserts as fact, and the judgment a careful human reader supplies tacitly, the kind of claim, its provenance, the binding status of the instrument, must therefore be supplied in structure or it is lost. This paper has set out a method that supplies it: three orthogonal axes whose intersection defines, mechanically, the subset that may be cited as current binding law; build-time gates, each demonstrated to bite, that make the subset and the discipline real rather than self-reported; a citation firewall that keeps machine assistance from manufacturing the citations the reference most needs to exclude; a computed-versus-authored reconciliation that treats disagreement as a finding and ships the derived layer as a labelled preview; and an honest-residual discipline that publishes the backlog as queryable data and insists that the gates are defenses, not a cure. The method is domain-general; the cross-border stablecoin register is its demonstration; and what carries to any other regulated field is the whole apparatus except the dimension set and the signal table, which the analyst of that field supplies. The claim the paper makes for the method is narrow and, it hopes, durable: a legal reference that is to be consumed by machines should be citable by construction, its citability a property the build guarantees and the consumer can rely on, and honest by construction about everything it has not yet verified.

References

> **A note on this bibliography, in the spirit of the paper.** The method argues that no tool should generate a citation and that every locator should be verified by a human against its source. This paper holds its own references to the same standard. The external entries below have been verified against their primary records, the publishers' and standards bodies' own pages (Springer, ACM, Oxford Academic, the W3C technical-reports index, and the OASIS standards registry), and each now carries its DOI or standard identifier and confirmed volume, pages, and date. In the vocabulary of the paper, these entries have been promoted from `firm_summary` to `resolution_text`: the bibliographic facts are taken from the authoritative source, not from memory. Entries for the author's own portfolio are self-published and carry their own persistent identifiers.

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- Fan, Y. Cross-Border Digital-Finance Corridor Atlas, v0.2.5 (working paper).
- Fan, Y. Cross-Border Stablecoin Feasibility Over Time, v0.1.0 (working paper): the companion forward-feasibility paper that applies this method's evidence model and labelled-preview discipline in the time dimension.

Companion works and instance. The method is demonstrated on the Cross-Border Stablecoin Register (CBSR, v0.9.8; data CC-BY-4.0, code Apache-2.0; schema-validated, archived to Zenodo for a DOI per release, queryable over the Model Context Protocol). The substantive legal facts it carries are analyzed in: Fan, Multi-Jurisdiction Stablecoin Compliance Matrix (v0.9.7); Fan, Cross-Border Stablecoin Architecture (working paper, v0.2.8); and Fan, Cross-Border Digital-Finance Corridor Atlas (v0.2.5). The claim-kind distinction originates in the Corridor Atlas's separation of primary-source legal constraints from market-reported operability. The most extended application of this method's computed-versus-authored reconciliation and its labelled-preview discipline, carried into the time dimension, where the alluded-to date-aware derivation of § 5.2 is developed in full, is Fan, Cross-Border Stablecoin Feasibility Over Time (working paper, v0.1.0), whose forward, trigger-conditioned feasibility map rests explicitly on the evidence model and the preview posture set out here.

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